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HW2- Problem 2

[-4, -3, -2, -1, 0, 1, 2, 3, 4, 5]

Left = 1

Right = 10

$M = 1+10 / 2 \rightarrow 5$

whatdoido(1, 5)

 whatdoido(1, 2)

 whatdoido(1, 1)

 A[1] = -4

 Return (0, 0, 0, -4)

 whatdoido(2, 2)

 A[2] = -3

 Return (0, 0, 0, -3)

Maxsum = max(0, 0, 0+0) $\rightarrow$ 0

Leftalignedmaxsum = max(0, -4+0) $\rightarrow$ 0

Rightalignedmaxsum = max(0, -3+0) $\rightarrow$ 0

Sum = -4+-3 $\rightarrow$ -7

(0, 0, 0, -7)

whatdoido(3, 5)

 whatdoido(3, 4)

 whatdoido(3, 3)

 (0, 0, 0, -2)

 whatdoido(4, 4)

 (0, 0, 0, -1)

 (0, 0, 0, -3)

 whatdoido(5, 5)

 (0, 0, 0, 0)

 (0, 0, 0, -3)

(0, 0, 0, -10)

whatdoido(6, 10)

 whatdoido(6, 8)

 whatdoido(6,7)

 whatdoido(6,6)

 A[6] = 1

 Return (1,1,1,1)

 whatdoido(7,7)

 A[7] = 2

 Return (2,2,2,2)

 Maxsum = (1, 2, 1+2) $\rightarrow$ 3

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    Leftalignedmaxsum = max(1, 1+2) -> 3
    Rightalignedmaxsum = max(2, 2+1)-> 3
    Sum= 1+2 ->3
    (3, 3, 3, 3)
    whatdoido(8,8)
        A[8] = 3
        Return (3,3,3,3)
    (6, 6, 6, 6)
    whatdoido(9, 10)
        whatdoido(9,9)
            A[9] = 4
            Return (4,4,4,4)
        whatdoido(10,10)
            A[10] = 5
            Return (5,5,5,5)
    (9, 9, 9, 9)
    (15,15,15,15)

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Maxsum = max(0, 15, -10+15) -> 15
Leftalignedmaxsum = max(0, -10+15) = 5
Rightalignedmaxsum = max(15, 0+15) = 15
Sum = 5

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Return (15, 5, 15, 5)

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a.)

$$T(n) = 2T(n/2) + cn$$

$$T(1) = c$$

b.)

$$T(n) = 2T(n/2) + cn$$

Substitute

$$\begin{aligned}
 T(n) &= 2(2T(n/4) + c(n/2)) + cn \\
 &= 4T(n/4) + 2c(n/2) + cn \\
 &= 4T(n/4) + (2/2 + 1)cn \\
 &= 4(4T(n/8) + c(n/4) + (2/2+1)cn) \\
 &= 16T(n/8) + 4c(n/4) + (2/2 + 1)cn \\
 &= 16T(n/8) + (4/4+2/2+1)cn
 \end{aligned}$$

$a=2, b=2, d=1, a/b^d = 2/2 = 1$, master theorem says $O(n^{\log a / \log b} \cdot \log n)$ if a/b^d is 1, which it is. Simplifies to $O(n^1 \log n) \rightarrow O(n \log n)$

c.) Sum is the sum of all integers in the array. Maxsum is the largest sum that can be made from the array. Leftalignedmaxsum is the biggest sum that can be made starting from the leftmost value of the array and moving right. Rightalignedmaxsum is the same, but starting from rightmost and moving left.